Collective AI can amplify tiny perturbations into divergent decisions

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Large language models are increasingly used in groups that discuss a problem and then vote or produce a joint recommendation. Group deliberation is often expected to make AI decisions more reliable. We find that it can instead make them less predictable. In a fully deterministic, self-hosted benchmark, exact reruns are identical, but small wording changes intended to preserve a scenario's meaning can grow over repeated discussion and often change the final recommendation. In deployed black-box API systems, nominally identical committee runs also vary at temperature 0, a setting many users expect to be nearly deterministic. Across 12 policy scenarios, the results show that instability is not only a consequence of deployment-side variation: repeated interaction can make a committee sensitive to small differences in how a case is presented. Additional deployed experiments show that role structure, model composition, and the amount of prior discussion agents can revisit all change the degree of divergence. Collective AI therefore presents a stability problem as well as an accuracy problem. Deterministic execution alone does not ensure that a committee's recommendation is predictable or readily auditable.

Large language models are increasingly deployed not only as individual assistants but as committees whose members exchange arguments and then vote or synthesize a recommendation (*1–5*). The appeal is straightforward: a group might correct the blind spots of any one model (*6, 7*). Yet a deliberating committee does more than pool independent opinions. Each contribution becomes part of the next round of discussion, creating a feedback loop. Such loops can cancel small differences, but they can also make them grow. For committees used in policy advice, institutional decision support, or other consequential settings, that possibility is a basic reliability concern.

We ask a simple question: does repeated AI deliberation smooth away small differences, or turn them into different decisions? Answering it requires two separate tests. The practical test asks whether deployed API committees vary across nominally identical runs, even when users set temperature to 0 and expect near-determinism. The more fundamental test asks whether a fully deterministic committee can still respond differently to nearby starting points, such as wording changes intended to preserve a scenario's meaning. If instability reflects only deployment-side variation, better infrastructure might largely remove it. If exact reruns match but nearby inputs still pull apart, then the vulnerability is also built into the repeated interaction of the committee.

Prior work has shown that LLM outputs can be sensitive to prompt design and formatting (*8*), and long-standing work in the social sciences has shown that collective judgments depend on framing, composition, and institutional structure (*9–11*). What remains unclear is whether instability in AI committees is mainly a deployment artifact or a more structural property of repeated interaction. We study five-agent committees over 20 rounds in two complementary settings. First, we construct a fully deterministic self-hosted benchmark in which exact reruns produce the same digital artifacts, while only the scenario wording is varied in meaning-conservative ways. Second, we hold prompts fixed and repeat deployed black-box API committees at temperature 0. In both settings, we track how far the committees' average stated preferences move apart over the course of deliberation.

Small wording changes can redirect a deterministic committee

The deterministic hosted benchmark provides the cleanest test of whether collective AI can amplify small changes after run-to-run variation has been removed. We used Qwen/Qwen2.5-7B-Instruct in a fixed single-GPU environment with greedy decoding, pinned seeds, and deterministic

CUDA/PyTorch settings. Exact reruns of the same 20-round committee pipeline were artifact-identical. We then changed only the scenario text, using 20 manually screened variants per condition: one canonical wording, 10 surface rephrasings, and 9 formatting-only changes. Committee instructions, role prompts, parser structure, voting logic, and execution settings were held fixed.

At every round, agents wrote an argument and recorded a structured preference among three policy options. We averaged those preferences within each committee and measured the average pairwise distance, $D(t)$, between committees over time. We summarize whether that distance tends to grow with a log-linear slope, $\hat{\lambda}$, estimated over rounds 3–20; positive values indicate that committees are separating over the discussion. In the deterministic benchmark, the comparison is across nearby wording variants; in the deployed benchmark, it is across nominal reruns at fixed settings.

Under these controlled perturbations, deterministic committees still moved apart over rounds (Fig. 1). In the health-financing scenario HL-01, no-role committees reached $\hat{\lambda}_{\text{pert}} = 0.072$ with decision fragility 0.60, whereas role-structured committees reached $\hat{\lambda}_{\text{pert}} = 0.048$ with fragility 0.35. Decision fragility is one minus the plurality share across variants; 0.60 means that the most common recommendation appeared in only 40% of nearby variants. Across the 12-scenario benchmark, nearby variants often changed the final decision and perturbation-divergence slopes were positive in almost all conditions. Exact rerun determinism therefore did not guarantee a robust decision: even semantically conservative changes to the initial scenario could redirect a committee's deliberation.

Deployed committees vary, and design choices matter

The deployed API experiments provide a practical counterpart to the hosted benchmark. Prompts and settings were fixed and each condition was rerun up to 20 times at temperature 0. This regime does not isolate sensitivity to a small input change as cleanly as the deterministic benchmark, because deployment-side variation remains possible. It does, however, represent the setting in which institutions are likely to use committee systems.

In HL-01, a uniform NoRoles committee had a positive but comparatively low divergence slope ($\hat{\lambda} = 0.0221$). Adding role mandates to the same model increased divergence ($\hat{\lambda} = 0.0541$), and a

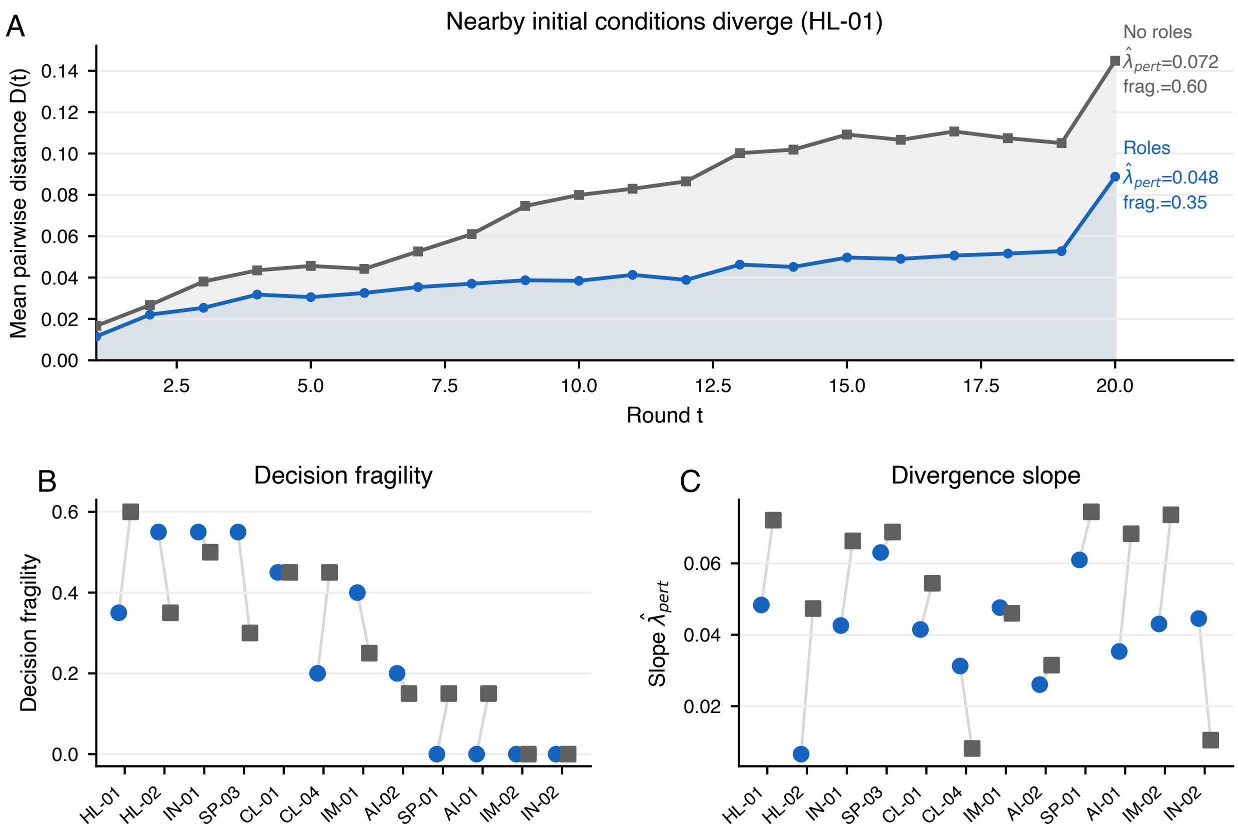

Figure 1: **Small wording changes can redirect deterministic committees.** Hosted runs were executed in a fixed deterministic environment, and exact reruns of the same full 20-round committee pipeline were artifact-identical. (A) Mean pairwise trajectory distance $D(t)$ in HL-01 under meaning-conservative perturbations to the scenario text, shown separately for Roles and NoRoles. (B) Cross-scenario decision fragility under deterministic perturbations, defined as one minus the plurality share across 20 variants. (C) Cross-scenario perturbation-divergence slope $\hat{\lambda}_{\text{pert}}$. The variation therefore reflects controlled changes to initial conditions rather than stochastic reruns.

mixed-model NoRoles committee was higher still ($\hat{\lambda}$ = 0.0947; Fig. 2A). The full 2×2 matrix also shows that these factors do not follow a simple additive rule: the mixed+roles condition was less unstable than mixed+no-role ($\hat{\lambda}$ = 0.0519 versus 0.0947; Fig. 2B). The displayed difference-in-differences estimate is local to HL-01 and is not evidence of a scenario-general interaction. Across the broader scenario set, uniform NoRoles was usually the lowest-divergence configuration, but it often remained in a positive-$\hat{\lambda}$ regime. Thus, temperature 0 should not be treated as a guarantee of reproducible collective behavior (Supplementary Fig. S8).

Role structure is one architectural channel, but not the only one. In role-structured deployed committees, removing the Chair mandate produced the largest reduction in HL-01 divergence among single-role ablations (Fig. 3A). Across five scenarios, the Chair effect was positive in four, with the strongest and best-resolved effects in IM-01 and HL-01 (Fig. 3B). This pattern does not make the Chair a universal source of instability. It suggests that a synthesis-focused role can be an important amplifier in this deployed, role-structured setting.

Shorter memory is associated with less separation

We next tested whether limiting access to earlier arguments changes the pattern. Reducing argument-memory depth from $k = 15$ to $k = 3$ lowered $\hat{\lambda}$ in all four tested scenarios (AI-01, CL-01, HL-01, and SP-03). A stricter memory-collapse variant ($k = 1$) also lowered $\hat{\lambda}$ relative to baseline Roles in IM-01 and CL-01 (Fig. 4). These tests do not eliminate divergence and do not isolate memory alone, because a shorter window also gives each agent less information. They nevertheless provide descriptive evidence that stability is partly a matter of deliberation protocol.

Implications for auditing collective AI

Multi-LLM committees are intended to make AI judgment more dependable: a group may correct the errors or blind spots of a single model. Our results show a limit to that intuition. Once agents repeatedly react to one another, the discussion becomes a feedback process. Small changes can grow into different lines of reasoning and sometimes different recommendations.

The deterministic benchmark makes the source of that problem clearer. Exact reruns of the full pipeline were artifact-identical, removing residual execution variation as an explanation. Yet

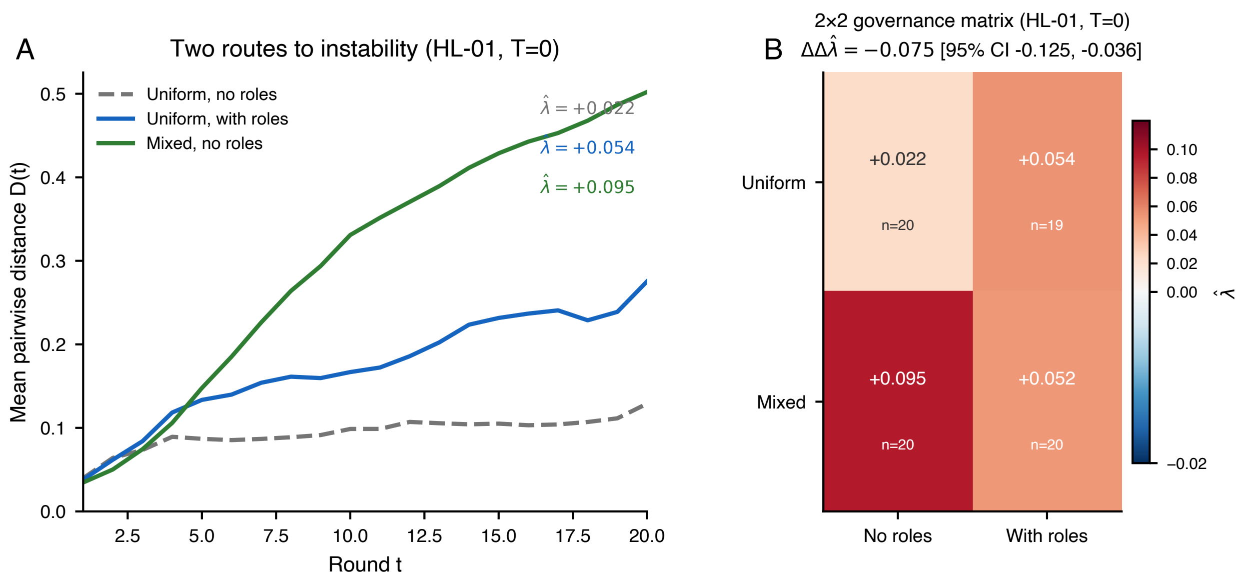

Figure 2: **Committee design changes deployed instability (HL-01, $T = 0$).** (A) Mean pairwise trajectory distance $D(t)$ for uniform NoRoles, uniform Roles, and mixed NoRoles committees. Adding roles in a uniform committee and using mixed models without roles both increase divergence. (B) The 2×2 matrix crosses role structure and model composition. The displayed difference-in-differences estimate and its 95% replicate-bootstrap confidence interval define the HL-01 non-additivity check. Realized sample sizes are shown in cell; uniform+roles has $n = 19$ and all other cells have $n = 20$.

wording changes that preserved the facts, options, and endpoint still led committees apart over time and often changed the recommendation. Repeatability, in other words, is not the same as robustness.

This matters for auditing and institutional use. A single transcript can be coherent and persuasive while failing to represent what the same system would recommend after an innocuous rephrasing. Auditing should therefore ask more than whether one run is well reasoned. It should test whether the recommendation survives nearby inputs that should not change the substance of the case. This focus on measured and documented risk aligns with AI risk-management guidance (*12*).

Our claims are empirical and deliberately bounded. The slope $\hat{\lambda}$ is an operational index of growing separation, not a formal proof of deterministic chaos (*13, 14*). We do not claim that every multi-LLM committee is unstable for every task, model, or protocol. We show instead that in this family of deliberative AI systems, instability can be induced experimentally, modified by protocol choices, and survive deterministic execution. The next step is to connect stability to consequences such as accuracy, calibration, welfare, and decision harm, and to develop protocols that limit unwanted amplification without eliminating useful disagreement.

Materials and methods

Protocol. We used a windowed-summary deliberation protocol for five-agent committees run over 20 rounds. At each round, every agent received the scenario packet, a committee state table, and a transcript window of prior arguments (default $k = 15$), then returned free-form argument text and a structured state line. A deterministic parser extracted preference vectors from that state line, and one automatic repair pass was allowed after malformed output.

Scenarios. The benchmark contains 12 structured policy scenarios spanning immigration, health, income, climate, speech, and AI governance (two scenarios per domain). Each scenario presents three discrete policy options and a fixed decision endpoint.

Models and conditions. The deterministic hosted benchmark used Qwen/Qwen2.5-7B-Instruct. The deployed API benchmark used GPT-4.1-mini for uniform committees. Mixed-model committees used Chair (GPT-4.1), Welfare (Claude Sonnet 4.6), Rights (Gemini 2.5 Flash), Equity (Grok-3-mini), and Security (GPT-4.1-mini). Core deployed conditions crossed NoRoles/Roles with uniform/mixed model composition at $T = 0$. Additional deployed runs covered role ablation and memory-window variants ($k = 3$, $k = 1$).

Deterministic hosted extension. The hosted benchmark used a fixed single-GPU environment (NVIDIA L4, bfloat16), greedy decoding, single-process execution, pinned seeds, and deterministic CUDA/PyTorch settings. Exact reruns of the same committee condition were artifact-identical at the full-pipeline level. Only the scenario text was perturbed through 20 manually screened, meaning-conservative variants; committee instructions, parser schema, voting logic, and execution settings were fixed.

Inference and uncertainty. For each condition, we computed the committee mean trajectory and $D(t)$. In the deterministic benchmark, $D(t)$ was computed across nearby prompt perturbations; in the API benchmark, it was computed across nominal reruns at fixed settings. We estimated $\hat{\lambda}$ from log-linear fits of $D(t)$ over rounds 3–20, and denote the hosted perturbation analogue by $\hat{\lambda}_{\text{pert}}$. Where reported, 95% confidence intervals were obtained from 500 bootstrap resamples of complete committee trajectories. For the HL-01 2×2 comparison, each cell was resampled independently. Realized sample size is reported at panel or cell level. Reporting follows reproducibility-oriented practice for empirical ML benchmarks (*15*).

Implementation. Experiments and analyses were implemented in Python 3.11 with NumPy, SciPy, and Matplotlib (*16, 17*).

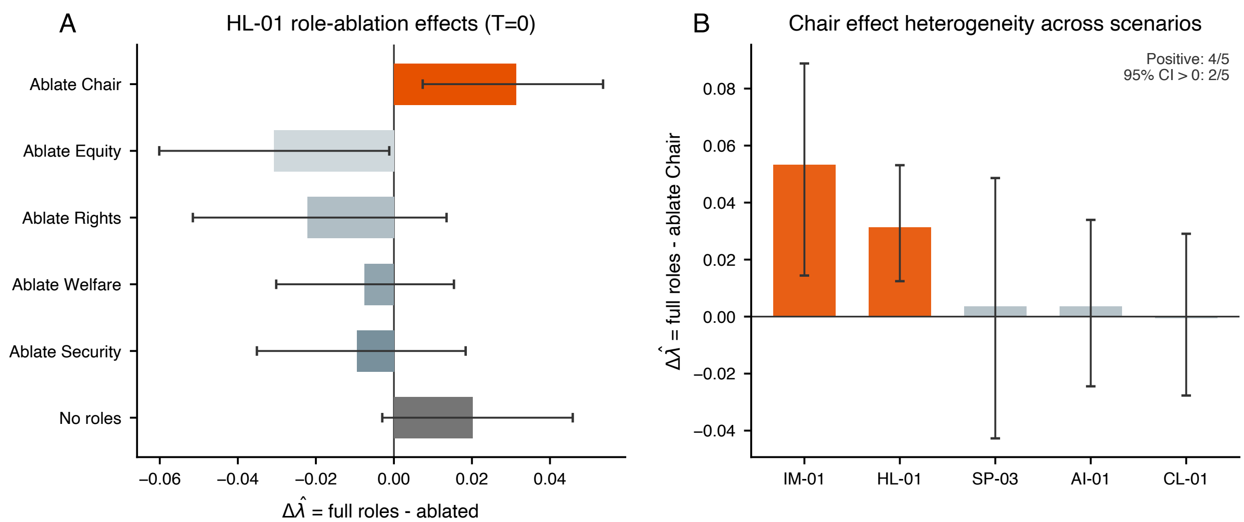

Figure 3: **The Chair role can amplify instability in role-structured deployed committees.** (A) HL-01 role-ablation effects on $\Delta\hat{\lambda} = \hat{\lambda}_{\text{full roles}} - \hat{\lambda}_{\text{ablated}}$. Chair ablation yields the largest reduction among role mandates, with a no-role reference shown for context. (B) Scenario-level Chair effects across IM-01, HL-01, SP-03, AI-01, and CL-01. Point estimates are positive in four of five scenarios, with strongest evidence in IM-01 and HL-01.

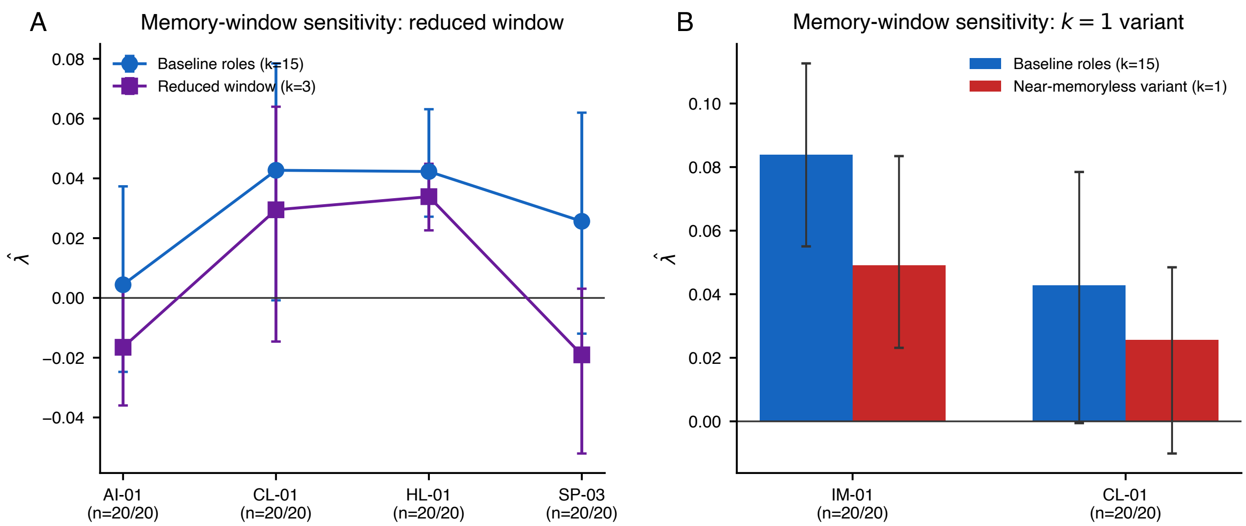

Figure 4: **Remembering less of the discussion is associated with less separation.** (A) Reducing the memory window from $k = 15$ to $k = 3$ lowers the point estimate of $\hat{\lambda}$ in all four displayed scenarios. (B) A $k = 1$ variant likewise has lower point estimates in IM-01 and CL-01 than the $k = 15$ baseline. Whiskers show 95% replicate-bootstrap confidence intervals and realized sample sizes are annotated in the panels. These descriptive interventions also change the information available to agents.

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Competing interests: The authors declare no competing interests.

Data and materials availability: The run artifacts, input materials, and analysis code supporting the findings are publicly available at `https://github.com/shimaohajime/Collective-AI-divergence`.

Supplementary materials

Materials and Methods S1 to S6

Supplementary Figures S1 to S8

Supplementary Tables S1 to S8